ELL 409 Course Project 2
High-Dimensional Clustering Under Heterogeneous Structure

Group 9

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Abstract

This report presents our end-to-end methodology for ELL 409 Course Project 2 on high-dimensional unsupervised clustering. The benchmark comprises four subproblems (Parts A–D): sparse marker clustering in high-dimensional noisy data (A), rare cell-type discovery under severe cluster imbalance (B), hierarchical and continuously transitioning cell-state clustering (C), and spatial tissue-niche inference using both molecular features and neighbourhood context (D). Performance is evaluated by Adjusted Rand Index (ARI), with final score $0.20\,\mathrm{ARI}_A + 0.25\,\mathrm{ARI}_B + 0.30\,\mathrm{ARI}_C + 0.25\,\mathrm{ARI}_D$.

All four parts share a common preprocessing backbone: variance-based gene filtering, median imputation, per-batch mean centering, RobustScaler normalisation, and PCA. Part-specific algorithms are then deployed: Gaussian Mixture Models with BIC-based cluster-count selection (A); UMAP followed by HDBSCAN with adaptive noise handling (B);multi-seed Spectral Clustering with seed-stability selection (C); and spatial-context-aware Ward agglomerative clustering with block standardisation and a three-dimensional hyperparameter sweep (D). The report covers EDA findings, mathematical derivations, preprocessing justifications, an internal validation protocol, a baseline comparison, rejected alternatives, an ablation study, and a part-wise metric breakdown. The final submission achieves a weighted ARI of **0.8823** on the held-out test set.

Contents

1 Exploratory Data Analysis and Preprocessing

1.1 Key EDA Observations

1. **Dataset dimensions.** The training set contains 6,600 rows and 581 columns. Part A has 1,400 rows, Part B has 1,600, Part C has 1,800, and Part D has 1,800. The test set contains 5,800 rows (A: 1,200; B: 1,400; C: 1,600; D: 1,600). The 581 columns comprise 500 gene-expression-like features (g1–g500), 40 protein features (p1–p40), 30 morphology features (morph1–morph30), three QC scalars, four spatial features, and metadata columns.
2. **Mode separation.** Parts A–C consist of `mode=single_cell` rows; Part D consists entirely of `mode=spatial_spot` rows. Spatial coordinate and neighbourhood-summary features are informative only for Part D and are excluded from single-cell feature sets.
3. **Missing-value pattern.** The training data contains 3,551 missing entries, confined to protein features (≈0.4–0.6%) and morphology features (≈0.8–1.3%) per part. Gene and QC features are complete. This sparse, structured missingness is handled by median imputation.
4. **Gene expression range and log-transformation.** The minimum observed gene value is negative (−6.19), making $\log(1 + x)$ transformation mathematically invalid. Empirical tests confirmed that enabling log1p caused performance regressions; the transformation is disabled throughout.
5. **Gene variance structure.** All 500 genes have positive variance in every part. Top-gene variances are ≈0.70–1.01 for Parts A–C but substantially lower at ≈0.43–0.47 for Part D, reflecting more compressed molecular signals in spatial tissue data. Variance filtering with a part-specific threshold captures the dominant signal efficiently.
6. **Batch effects.** Five distinct batch IDs are present. Without correction, batch-to-batch mean differences dominate PCA components and corrupt clustering. Per-batch mean centering is an essential preprocessing step for all parts.
7. **No ground-truth labels.** Neither split contains `cluster_id`. All model selection relies on internal clustering metrics and clustering stability. A key empirical observation is that silhouette score does *not* reliably predict ARI on the held-out test set, as discussed in Sections 2 and 8.

1.2 Preprocessing Pipeline

The same five-stage pipeline is applied to all parts. All statistics are fitted on the training split and applied to the test split without refitting.

1.2.1 Variance-Based Gene Filtering

Let $\hat{v}_j = \mathrm{Var}_{\mathrm{train}}(g_j)$ be the training variance of gene j. The selected gene set is

$$\mathcal{G}_{\mathrm{sel}} = \Big\{ j : \mathrm{rank}(-\hat{v}_j) \leq N \Big\},$$

where N is part-specific. The full base feature set is $\mathcal{G}_{\mathrm{sel}} \cup \{p_1, \ldots, p_{40}\} \cup \{\mathrm{morph}_1, \ldots, \mathrm{morph}_{30}\} \cup$ QC.

Part	Top Genes (N)	Proteins	Morphs + QC	Total
A	180	40	33	253
B	220	40	33	293
C	240	40	33	313
D	180	40	33	253

Table 1: Feature counts per part after variance-based filtering.

1.2.2 Median Imputation

Let $\tilde{x}_j$ be the training median of column j. For every missing entry x_{ij}, we substitute $x_{ij} \leftarrow \tilde{x}_j$. The median is preferred over the mean because its breakdown point is 50%, making it resistant to extreme values in protein and morphology features.

1.2.3 Batch Effect Correction by Per-Batch Mean Centering

Let $b(i) \in \{1, \ldots, 5\}$ denote the batch of observation i and $\mu_{j,b} = \mathrm{mean}_{i:\, b(i)=b}(x_{ij})$. The corrected feature is

$$x'_{ij} = x_{ij} - \mu_{j,\, b(i)}.$$

This removes additive batch offsets, ensuring PCA components reflect biological variation rather than measurement artefacts.

1.2.4 Robust Scaling

Each feature is scaled by the interquartile range:

$$x''_{ij} = \frac{x'_{ij} - Q_{2,j}}{Q_{3,j} - Q_{1,j}},$$

where $Q_{1,j}$, $Q_{2,j}$, $Q_{3,j}$ are the 25th, 50th, and 75th training percentiles. RobustScaler is preferred over StandardScaler because its IQR denominator is insensitive to the extreme marker values common in biological data.

1.2.5 PCA Dimensionality Reduction

The scaled matrix is projected onto the leading principal components:

$$Z = X_{\text{scaled}} V_d,$$

where $V_d \in \mathbb{R}^{p\times d}$ contains the top-d eigenvectors of the sample covariance matrix. Part-specific d: A = 30, B = 35, C = 40, D = 30. PCA concentrates cluster-separating variance, removes residual noise dimensions, and reduces the cost of downstream algorithms.

2 Cluster-Count Estimation and Validation Protocol

Since ground-truth labels are withheld, model selection relies on two complementary signals.

2.1 Internal Clustering Metrics

Three standard internal metrics are computed on training predictions:

- **Silhouette score:** $S = \frac{1}{n}\sum_i \frac{b_i - a_i}{\max(a_i, b_i)}$, where a_i is the mean intra-cluster distance and b_i is the mean distance to the nearest other cluster. $S \in [-1, 1]$; higher is better.
- **Calinski–Harabasz index:** $\mathrm{CH} = \frac{SS_B/(K-1)}{SS_W/(N-K)}$, where SS_B and SS_W are the between- and within-cluster sums of squares. Higher is better.
- **Davies–Bouldin index:** $\mathrm{DB} = \frac{1}{K}\sum_k \max_{j\neq k} \frac{s_k + s_j}{d(c_k, c_j)}$, where s_k is mean intra-cluster distance and $d(c_k, c_j)$ is the centroid distance. Lower is better.

Critical caveat. Silhouette does not reliably predict ARI. For Part C, a competing pipeline achieved silhouette 0.606 versus 0.096 for the chosen Spectral solution, yet SpectralClustering produced substantially higher ARI on the held-out set (Section 8). Stability is therefore the primary selection criterion.

2.2 Stability-Based Selection

Multi-seed spectral stability (Part C). SpectralClustering is run with $n_s = 5$ random seeds. Stability is the mean pairwise ARI across all $\binom{5}{2} = 10$ pairs:

$$\text{stability}(k, nn) = \frac{2}{n_s(n_s - 1)} \sum_{i<j} \mathrm{ARI}(\mathbf{L}_i, \mathbf{L}_j).$$

Configurations with stability ≥ 0.5 are retained; among these the configuration with highest stability (silhouette as tie-break) is selected.

Bootstrap subsample stability (Part D). Two independent 90% subsamples S_a and S_b are drawn. For each configuration $\theta = (K, \alpha, \beta)$, the pipeline is fitted on each subsample and stability is computed on the overlap $S_a \cap S_b$:

$$\text{stability}(\theta) = \text{ARI}\Big(\mathbf{L}_a[\text{overlap}],\, \mathbf{L}_b[\text{overlap}]\Big).$$

Configurations with stability ≥ 0.5 are retained; selection by highest stability, tie-broken by silhouette.

3 Baseline Model

A global K-means baseline is established: for each part, K is set to the recommended midpoint and clustering is applied to the raw feature matrix (no filtering, no PCA). K-means minimises

$$\min_{\mu_1,\ldots,\mu_K} \sum_i \min_k \|x_i - \mu_k\|^2$$

via Lloyd's alternating assignment-and-update algorithm. Failure modes:

- **Part A.** The 320+ non-selected noisy genes dominate distances; centroids aggregate noise rather than marker signal.
- **Part B.** Equal-size centroid assumptions render rare groups (∼20–50 observations) invisible against dominant clusters of ∼500.
- **Part C.** Flat Voronoi boundaries cannot capture hierarchical and manifold-like transitions between cell states.
- **Part D.** Each row is treated as independent; spatial context is ignored entirely.

These structural failures motivate the part-specific pipelines below.

4 Part-Specific Models: Methodology and Mathematics

4.1 Part A: Gaussian Mixture Model with BIC-Based K Selection

4.1.1 Intuition

After PCA compression, marker-driven populations form approximately Gaussian clouds in the reduced space. A GMM uses soft assignments and per-cluster covariance, capturing elongated shapes that K-means cannot. Diagonal covariance is chosen over full because at $d = 30$ components and cluster sizes of ∼100–300, full covariance ($d(d+1)/2 = 465$

parameters per component) is prone to ill-conditioning; diagonal ($d = 30$ parameters) is a principled regularisation consistent with the approximate independence of PCA components within each cluster.

4.1.2 Mathematical Formulation

A K-component GMM models the data density as:

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x;\, \mu_k, \Sigma_k), \qquad \sum_k \pi_k = 1,$$

with $\Sigma_k = \mathrm{diag}(\sigma^2_{k1}, \ldots, \sigma^2_{kd})$. Parameters are estimated by the EM algorithm:

E-step:

$$r_{ik} = \frac{\pi_k \mathcal{N}(x_i;\, \mu_k, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(x_i;\, \mu_j, \Sigma_j)}.$$

M-step:

$$N_k = \sum_i r_{ik}, \quad \pi_k = \frac{N_k}{n}, \quad \mu_k = \frac{1}{N_k} \sum_i r_{ik}\, x_i, \quad \sigma^2_{kj} = \frac{1}{N_k} \sum_i r_{ik}(x_{ij} - \mu_{kj})^2.$$

Ten random initialisations and regularisation $\varepsilon = 10^{-4}$ on diagonal entries are used.

4.1.3 BIC-Based K Selection

We sweep $K \in \{7, 8, 9, 10, 11, 12, 13\}$ and select the value minimising:

$$\mathrm{BIC}(K) = -2\,\ell(X; \hat{\theta}_K) + p_K \ln n, \qquad p_K = K(2d + 1) - 1.$$

The penalty $p_K \ln n$ accounts for K mean vectors (each of length d), K diagonal variance vectors, and $K - 1$ free mixing weights.

4.1.4 Preprocessing (Part A)

Top-180 variable genes $\cup$ proteins, morphs, QC $\rightarrow$ median imputation, batch centering, RobustScaler $\rightarrow$ PCA(30) $\rightarrow$ BIC sweep, final GMM with 10 restarts.

4.2 Part B: UMAP + HDBSCAN with Adaptive Noise Handling

4.2.1 Intuition

Detecting rare clusters ($\sim$20–50 observations) alongside dominant clusters of $\sim$500 requires a method that does not assume equal or roughly similar cluster sizes. HDBSCAN builds a full density hierarchy and extracts clusters at multiple density levels simultaneously, making it naturally robust to imbalance. UMAP is applied first to compress the 35-dimensional PCA embedding into a 10-dimensional manifold-preserving representation, sharpening the local density structure that HDBSCAN exploits.

4.2.2 UMAP Embedding

The k-NN graph edge weight in high-dimensional space is:

$$w_{ij} = \exp\left(-\frac{d(x_i, x_j) - \rho_i}{\sigma_i}\right),$$

where ρ_i is the nearest-neighbour distance of x_i and σ_i is a bandwidth such that $\sum_j w_{ij} = \log_2 k$. The low-dimensional embedding minimises the cross-entropy between high- and low-dimensional fuzzy simplicial sets:

$$\mathcal{L} = \sum_{i,j} \left[w_{ij} \log \frac{w_{ij}}{v_{ij}} + (1 - w_{ij}) \log \frac{1 - w_{ij}}{1 - v_{ij}} \right].$$

Parameters: $n_{\text{neighbours}} = 15$, min_dist $= 0.0$ (collapses same-neighbourhood points to sharpen cluster separation), $n_{\text{components}} = 10$.

4.2.3 HDBSCAN Clustering

Given the UMAP embedding, HDBSCAN computes the mutual reachability distance:

$$d_{\text{mrd}}(x_i, x_j) = \max\Big(\text{core}_k(x_i),\ \text{core}_k(x_j),\ d(x_i, x_j)\Big),$$

where $\text{core}_k(x)$ is the k-th nearest-neighbour distance of x. A minimum spanning tree on the mutual reachability graph yields a cluster hierarchy; the Excess-of-Mass (EOM) method extracts the flat clustering maximising total stability $\sum_c \lambda(c)$, where

$$\lambda(c) = \sum_{x \in c} \Big(\lambda_x - \lambda_{\text{birth}}(c)\Big), \qquad \lambda_x = 1/d_x.$$

This allows clusters to be retained at different density levels simultaneously.

4.2.4 Hyperparameter Sweep and Noise Handling

We sweep min_cluster_size $\in \{5, 6, 8, 10, 12, 15\}$ and min_samples $\in \{3, 5, 8\}$ (18 configurations). Selection criterion: $K_{\text{dense}} \in [5, 19]$, then minimise noise fraction, then minimise smallest cluster size. Noise points (label $= -1$) are handled adaptively: if noise fraction $< 5\%$ or count $< 2 \times$ min_cluster_size, assign all noise to one extra cluster; otherwise split noise with K-means.

4.2.5 Preprocessing (Part B)

Top-220 genes ∪ proteins, morphs, QC → median imputation, batch centering, RobustScaler, PCA(35) → UMAP(10 dims) → HDBSCAN(EOM, hyperparameters from sweep), adaptive noise handling.

4.3 Part C: Spectral Clustering with Multi-Seed Stability

4.3.1 Intuition

Part C contains hierarchical and continuously transitioning cell states. True clusters are connected along manifold-like trajectories rather than forming compact Euclidean blobs. Spectral Clustering partitions the spectrum of a k-NN affinity graph, capturing geodesic-like proximity along the manifold and separating states that are far apart along the manifold even when nearby in the ambient PCA space.

4.3.2 Mathematical Formulation

From PCA embedding Z, we build a binary k-NN affinity matrix A (where $A_{ij} = 1$ iff x_j is among x_i's k nearest neighbours or vice versa) and compute the normalised graph Laplacian:

$$L = I - D^{-1/2}\, A\, D^{-1/2},$$

where $D = \mathrm{diag}(A\mathbf{1})$ is the degree matrix. The spectral embedding $V_K \in \mathbb{R}^{n \times K}$ collects the K eigenvectors of L with smallest eigenvalues; K-means is then applied to the rows of V_K. Each eigenvector is a smooth function on the graph nodes, and together the first K eigenvectors define a space where graph-connected components are well-separated.

4.3.3 Grid Sweep and 5-Seed Stability

We sweep $k \in \{12, 14, 16, 18\}$ and $n_{\text{neighbours}} \in \{15, 20, 30\}$, running 5 seeds per configuration (Section 2). The configuration with highest stability ≥ 0.5 is selected. At inference, 3 independent runs are performed and the one with the highest silhouette score is returned to reduce random-initialisation variance.

4.3.4 Preprocessing (Part C)

Top-240 genes ∪ proteins, morphs, QC → median imputation, batch centering, RobustScaler, PCA(40) → SpectralClustering(k-NN affinity, hyperparameters from stability sweep).

4.4 Part D: Spatial-Context-Aware Ward Clustering

4.4.1 Intuition

Part D tissue-spot clusters (niches) depend on both a spot's own molecular profile and its spatial neighbourhood. Two spots with identical molecular features but different spatial environments may belong to different niches. We augment the molecular PCA embedding with k-NN neighbourhood statistics and raw spatial features, then apply deterministic Ward agglomerative clustering on the augmented representation.

4.4.2 Neighbourhood Feature Construction

Let $Z \in \mathbb{R}^{n\times d}$ be the PCA molecular embedding and let coordinates $C \in \mathbb{R}^{n\times 2}$ be $(x_{\text{coord}}, y_{\text{coord}})$. Build a binary k-NN connectivity graph A ($k = 8$ spatial neighbours):

$$\mathrm{NM}_i = \frac{1}{D_i}\sum_j A_{ij}\, Z_j, \qquad \mathrm{NS}_i = \sqrt{\frac{\sum_j A_{ij}\, Z_j^{\odot 2}}{D_i} - \mathrm{NM}_i^{\odot 2}},$$

where $D_i = \sum_j A_{ij}$ is the degree of node i and $\odot 2$ denotes element-wise squaring. NM captures the mean molecular identity of a spot's spatial neighbourhood; NS captures neighbourhood heterogeneity (spots at a niche boundary have high NS).

4.4.3 Block Standardisation

The molecular block Z, neighbourhood blocks NM and NS, and spatial block $\mathrm{SP} = (x_{\text{coord}}, y_{\text{coord}}, \text{neighbor\_density}, \text{local\_heterogeneity})$ have different natural scales. Without normalisation the block with the largest scale dominates Euclidean distance and the weight parameters α, β are uninterpretable. Block standardisation makes each block unit-RMS:

$$\texttt{standardize}(X) = \frac{X - \bar{X}}{\mathrm{RMS}(X - \bar{X})}, \qquad \mathrm{RMS}(Y) = \sqrt{\frac{1}{nd}\sum_{ij} Y_{ij}^2}.$$

The augmented embedding is then:

$$Z_{\text{final}} = \left[Z_{\text{blk}} \;\middle|\; \alpha\, \mathrm{NM}_{\text{blk}} \;\middle|\; \tfrac{\alpha}{2}\, \mathrm{NS}_{\text{blk}} \;\middle|\; \beta\, \mathrm{SP}_{\text{blk}} \right],$$

where α is the neighbourhood mean weight and β is the spatial coordinate weight. Locking the std-weight to $\alpha/2$ preserves a fixed mean-to-std ratio while reducing the sweep to three free parameters.

4.4.4 Ward Agglomerative Clustering

Ward's method merges the pair of clusters (A, B) minimising the increase in total within-cluster sum of squares:

$$\Delta(A, B) = \frac{|A|\,|B|}{|A| + |B|}\,\|\mu_A - \mu_B\|^2.$$

Ward is deterministic (no random initialisation), which makes it ideal for stability sweeps and produces compact, well-connected clusters.

4.4.5 Three-Dimensional Hyperparameter Sweep

Three parameters jointly determine cluster quality: K (number of clusters), α (neighbourhood mean weight), and β (spatial coordinate weight). We perform a full 3D grid sweep:

Parameter	Search grid	Count
K	$\{8, 10, 12, 14, 16\}$	5
α (smooth_weight_mean)	$\{0.1, 0.2, 0.3, 0.5\}$	4
β (spatial_coord_weight)	$\{0.25, 0.5, 1.0\}$	3

Table 2: Three-dimensional sweep grid for Part D. Total: $5 \times 4 \times 3 = 60$ configurations $\times$ 2 bootstrap subsets = 120 Ward fits.

Bootstrap stability (Section 2) selects the best configuration. The previously established default (K=10, α=0.3, β=0.5) is contained within the grid, so the worst case is a no-change outcome with no risk of regression.

4.4.6 Preprocessing (Part D)

Top-180 genes $\cup$ proteins, morphs, QC $\rightarrow$ median imputation, batch centering, RobustScaler, PCA(30) $\rightarrow Z$
k-NN graph ($k = 8$) on coordinates $\rightarrow$ compute NM, NS
Block standardise all four blocks; concatenate with sweep-selected (α, β)
Ward agglomerative clustering with sweep-selected K.

5 Models Considered but Not Adopted

The table below summarises alternatives evaluated during development and the reason each was rejected.

Alternative	Applied to	Reason for rejection
Leiden community detection	Part C	Requires tuning a resolution parameter γ without ground-truth labels; produced degenerate single-cluster outputs in all tested configurations.
UMAP + Ward clustering	Part C	Achieved silhouette 0.606 (vs. 0.096 for Spectral), yet produced lower ARI on the held-out set. Higher geometric compactness in UMAP space does not imply better recovery of graph-connected biological structure.
Ward hierarchical clustering	Part A	Deterministic and efficient, but GMM's soft EM assignments are better suited to clusters near shared boundaries; BIC further automates K selection.
Gaussian spatial smoothing	Part D	Gaussian-weighted k-NN neighbourhood statistics showed no improvement over binary k-NN connectivity while adding an extra bandwidth hyperparameter.
DBSCAN (without UMAP)	Part B	A global ε threshold performs poorly under variable cluster density; HDBSCAN on the UMAP manifold is substantially more effective at recovering rare groups.
Nearest-neighbour noise reassignment	Part B	Forcing all noise points into existing clusters discards potentially valid low-density structure in the UMAP embedding; adaptive noise handling preserves it.

Table 3: Alternatives considered and reasons for rejection.

6 Ablation Study

Ablation	Condition	Observation
Gene filtering	All 500 genes vs. top-N	Filtering improves silhouette and stability across all parts; unfiltered genes introduce noise dimensions that distort PCA-space distances.
Batch centering	With vs. without	Without centering, spectral embeddings (Part C) and PCA spaces (all parts) separate primarily by batch rather than by biological structure.
Log1p transformation	Enabled vs. disabled	Enabling log1p caused ARI regressions; minimum gene value is negative, making the transformation invalid. Disabled throughout.
Spectral stability (Part C)	2-seed vs. 5-seed mean pairwise ARI	The 2-seed estimate is noisy; 5-seed mean over 10 pairs is substantially more reliable and stops selecting configurations that agreed by chance.
Block standardisation (Part D)	With vs. without	Without block standardisation, α and β weight parameters are uninterpretable due to scale differences of $\sim 10\times$ across blocks; the 3D sweep is only meaningful after standardisation is applied.
Hyperparameter sweep (Part D)	1D (K only) vs. 3D (K, α, β)	The three parameters interact: a larger β may require a different K than a smaller β. Joint search finds better combinations than sequential 1D sweeps, with no risk of regression since the prior best config is inside the grid.

Table 4: Summary of ablation studies. All evaluations on the training split.

7 Part-Wise Metric Breakdown and Results

Since ground-truth labels are withheld, we report internal clustering metrics on the training split for the final pipeline.

Part	Algorithm	K selected	Silhouette	Davies-Bouldin	C-H Index
A	GMM (diag, BIC)	7–13	≈0.11	≈1.82	≈320
B	UMAP + HDBSCAN	6–20	≈0.20	≈1.65	≈180
C	SpectralClustering	12–18	≈0.10	≈2.10	≈95
D	Ward (augmented)	8–16	≈0.18	≈1.78	≈210

Table 5: Training-set internal metrics under the final pipeline. Ranges for K reflect the space explored by each part's selection procedure. Metric values are approximate and reflect variability across data subsets.

Note on Part C silhouette. The low silhouette (≈0.10) for Part C is an expected artefact of the manifold structure of the data, not a sign of poor clustering. SpectralClustering recovers graph-connected components that are geometrically diffuse in PCA space. Competing methods with higher silhouette produced lower ARI on the leaderboard (Section 8), confirming that seed stability is the correct selection criterion for this part.

The final submission achieves a weighted ARI of **0.8823** on the held-out Kaggle test set.

8 Discussion

8.1 Why Part-Specific Pipelines?

The four parts arise from fundamentally different data-generating mechanisms. Part A requires noise suppression and probabilistic soft-assignment. Part B requires density-based discovery of rare structures without equal-size assumptions. Part C requires graph-based spectral embedding to capture manifold topology. Part D requires spatial context integration. No single algorithm addresses all four requirements simultaneously, and a global pipeline must compromise across all four structural regimes.

8.2 The Silhouette Paradox for Part C

The most important empirical lesson from this project is that silhouette score can actively mislead algorithm selection in the presence of manifold structure. Silhouette measures how compact and well-separated clusters are in the ambient feature space. When true cluster boundaries follow a graph-connectivity topology rather than Euclidean hyperplanes, a partition that is compact in feature space may fragment or merge meaningful manifold-connected components. For Part C, SpectralClustering's graph-partition boundaries align with the true biological partition even though its clusters appear geometrically diffuse in PCA space. The correct signal for Part C model selection is therefore the multi-seed stability of SpectralClustering's own output, not silhouette.

8.3 Block Standardisation as an Architectural Requirement

The introduction of block standardisation was not a minor tuning step but an architectural correction. Without it, the smooth-weight α and spatial-weight β in Part D are pseudo-parameters: any chosen value is effectively overridden by the natural scale difference between blocks ($\sim 10\times$ difference in RMS). After block standardisation, each block has unit RMS and $\alpha = 0.3$ genuinely contributes 30% of the molecular block's effective scale. The 3D sweep is only interpretable after this correction is in place.

8.4 Strengths and Limitations

Strengths. All hyperparameters are selected by data-driven criteria (BIC, stability-based selection, silhouette tie-break) rather than by hand. The 3D Part D sweep is risk-free by construction. Stability-based selection guards against over-optimising internal metrics that do not predict ARI.

Limitations. Without ground-truth labels all model selection is indirect; a small labelled validation set would dramatically improve selection quality. UMAP's stochastic optimisation introduces seed sensitivity in Part B; multiple UMAP runs with majority-vote assignment could improve robustness. PCA dimensions are heuristically fixed; an explained-variance threshold could provide a more principled choice.

9 Conclusion

We presented a part-specific clustering framework for the ELL 409 Course Project 2 benchmark. Starting from a global K-means baseline that fails on all four parts for distinct structural reasons, we designed targeted solutions:

1. **Part A.** Diagonal GMM with BIC-based K selection over $\{7, \ldots, 13\}$ on a 30-dimensional PCA embedding of the top-180 variable genes. BIC automates K without an arbitrary hand-set value; diagonal covariance is well-regularised for the available sample sizes.
2. **Part B.** UMAP (n_neighbours=15, min_dist=0.0, 10 components) followed by HDB-

SCAN (EOM, hyperparameters from sweep). Adaptive noise handling detects rare subpopulations that centroid-based methods consistently absorb into dominant clusters.

3. **Part C.** SpectralClustering with k-NN affinity on a 40-dimensional PCA embedding, with $(k, n_{\text{neighbours}})$ chosen by a 5-seed stability sweep. Stability proved more reliable than silhouette; a UMAP+Ward alternative with $6\times$ higher silhouette produced lower ARI, confirming the inadequacy of silhouette for manifold-structured data.

4. **Part D.** Ward agglomerative clustering on a block-standardised augmented embedding combining molecular PCA, k-NN neighbourhood mean and standard deviation statistics, and raw spatial features. Block standardisation is an architectural prerequisite for the 3D hyperparameter sweep over (K, α, β), making the weight parameters interpretable and enabling principled joint search.

Key cross-cutting lessons: (1) silhouette score is an unreliable model-selection criterion when true cluster geometry is manifold-like; (2) block standardisation is essential before concatenating multi-source feature blocks with tunable weights; (3) stability-based selection is the most reliable unsupervised proxy for ARI when ground truth is unavailable; (4) algorithm choice matters more than hyperparameter tuning—replacing a structurally appropriate algorithm consistently led to large ARI regressions. The final pipeline achieves a weighted ARI of $\mathbf{0.8823}$ on the held-out test set.